

Advanced (Habanero)

Q1. Solve the inequality $|x| < 3$.

(A) $(-3, 3)$

(B) $(-\infty, -3) \cup (3, \infty)$

(C) $(-\infty, -3) \cup (3, \infty)$

(D) $[-3, 3]$

(E) $(-\infty, -3] \cup [3, \infty)$

Q2. The solution to $|2x - 1| \geq 5$ is

(A) $(-\infty, -2] \cup [3, \infty)$

(B) $(-\infty, -2) \cup (3, \infty)$

(C) $[-2, 3]$

(D) $[-3, -2] \cup [3, \infty)$

(E) $(-2, 3)$

Q3. Graph the solution set of $|x + 2| < 4$ on a number line.

(A) $(-6, 2)$

(B) $[-6, 2]$

(C) $(-2, 6)$

(D) $[-2, 6]$

(E) $(-\infty, -6) \cup (2, \infty)$

Q4. The absolute value inequality $|x - 4| > 2$ is equivalent to

(A) $x < 2$ or $x > 6$

(B) $x \leq 2$ and $x \geq 6$

(C) $x > 2$ and $x < 6$

(D) $x < 2$ and $x > 6$

(E) $x \geq 2$ or $x \leq 6$

Q5. Solve $|3x + 1| \leq 2$.

(A) $[-1, 1]$

(B) $\left[-\frac{3}{3}, \frac{1}{3}\right]$

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Q6. The solution set of $|x - 2| \geq 1$ is

(A) $(-\infty, 1] \cup [3, \infty)$

(B) $(-\infty, 1) \cup (3, \infty)$

(C) $[1, 3]$

(D) $(-\infty, 3] \cup [1, \infty)$

(E) $[1, 3]$

Q7. The inequality $|2x + 5| < 3$ is equivalent to

(A) $-\frac{8}{2} < x < -\frac{2}{2}$

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(A) $-\frac{8}{2} \leq x \leq -\frac{2}{2}$

$-\frac{8}{2} \leq x \leq -\frac{2}{2}$

(A) $-\frac{4}{2} < x < -\frac{8}{2}$

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Q8. Solve $|x^2 - 4| > 3$.

(A) $(-\infty, -1) \cup (1, \infty)$

(B) $(-\infty, -1] \cup [1, \infty)$

(C) $(-\infty, -2) \cup (2, \infty)$

(D) $(-\infty, 2) \cup (2, \infty)$

(E) $(-2, 2)$

Open Ended Questions (FRQ)

Q9. Model the situation 'the temperature must be within 5°F of 72°F ' using an absolute value inequality and solve for the possible temperatures.

Q10. A company packages nuts in bags that have a weight of 16 ounces with an allowed deviation of at most 1 ounce. Express this situation as an absolute value inequality and find the range of acceptable weights.

Chef's Tips

- Read each question carefully before selecting an answer.
- Show all work and explain reasoning for free response questions.
- Use a number line to help visualize the solution sets when necessary.